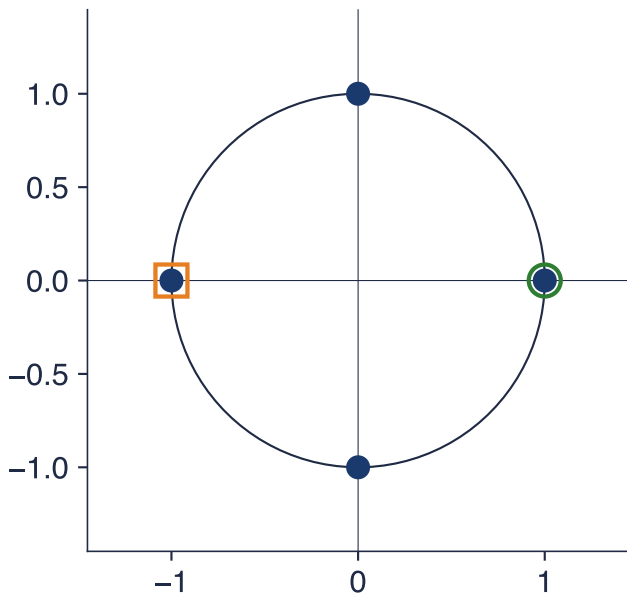
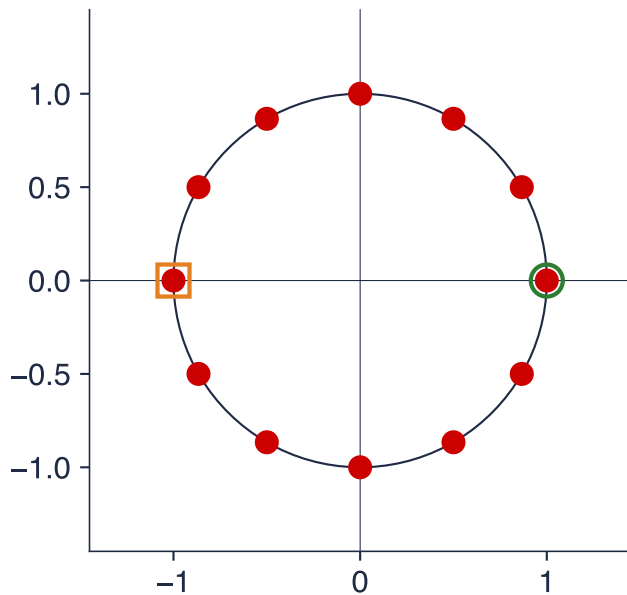


$s = 4$: 4 roots of $1 - z^4$



- roots of $1 - z^4$
- $z = 1$: zero frequency (trend)

$s = 12$: 12 roots of $1 - z^{12}$



- $z = -1$: frequency 1/2 (period 2)
- roots of $1 - z^{12}$